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Towards Accessible Legal AI: A Hierarchical Transformer-Based Approach for Summarizing Long Indian Legal and Policy Documents

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***Abstract*—** Policy and governmental documents tend to be quite extensive and contain complex legal language, which makes them less accessible to people. The existing transformer-based summarization approaches are intended for short documents and face difficulties in processing long legal and governmental documents in a readable manner. Summarization techniques, such as BART, PEGASUS, and T5, have achieved great success in abstractive summarization, but their input sequence lengths' limitations and lack of hierarchical modeling make their application to legal and long governmental documents impossible. This paper presents SET (Summarization for Enhanced Transparency), which is a hierarchical transformer-based approach to summarizing Indian legal and governmental documents that includes (i) sentence selection based on TF-IDF, (ii) abstractive summarization at the level of chunks based on BART-large-CNN, (iii) deduplication and filtering and (iv) the accessibility module for simplifying legal language. The hierarchical architecture allows performing the processing on CPUs and summarization of the documents with up to 200,000 words in length by splitting into chunks. Tested on a curated dataset with 77 documents of legal and governmental domain (≈2.78M words), SET demonstrates the ROUGE-2 score of 0.102, which surpasses baselines, while improving readability from Flesch-Kincaid grade of 15-25 to 50-60.

***Index Terms*—** *Text Summarization, Legal NLP, BART, Hierarchical Models, Accessibility*

I. INTRODUCTION

Legal documents play an important role in policy making and regulation, yet they tend to be lengthy, highly structured and use complicated legal jargon. Policy and legislative documents often feature abundant referencing and technical terms, thus there is a noticeable discrepancy between the availability and readability of legal documents [1], [2].

While transformer architectures like BART [3], PEGASUS [4], and T5 [5] have made significant progress in abstractive summarization, they have shown limitations when applied to legal documents due to their excessive length, poor hierarchical structuring, and a domain mismatch with the training corpora [6] – [8]. Existing solutions do not usually tackle readability, hence their outputs often remain hard for non-experts. In this paper, we present SET, which is a hierarchical system where sentences are initially extracted through TF-IDF [9], then the document is chunked for abstract summarization using BART-large-CNN, followed by applying the accessibility layer which transforms complex legal language to readable language without losing the meaning.The proposed approach was evaluated on a curated dataset consisting of 77 documents (≈2.78 million words) of legal and policy documents; it has shown improvement in summary quality and readability with a comparable ROUGE score, demonstrating the efficacy of hierarchical processing and readability improvement in enhancing public access to legal documents.

A.Contributions

- Hierarchical summarization approach using transformer models.
- Indian legal and governmental document corpus for long-document summarization.
- Accessibility layer to improve readability without losing information.
- Evaluation using ROUGE metrics and human evaluation.

II. RELATED WORK

Longformer utilizes sparse local/global attention mechanism for efficient handling of longer sequences [6]; BigBird incorporates local, random, and global attention mechanisms for achieving a similar performance [7]; LongT5 makes use of efficient attention in a text-to-text setting [8]; and in addition, Huang et al. [2] have presented Hepos, an efficient encoder-decoder attention mechanism, as well as the GovReport dataset for long-document summarization of government reports. These architectural modifications facilitate the processing of longer contexts but view documents as continuous sequences of tokens without preserving any hierarchical structure (sections,

sub-sections, clauses), and do not solve the problem of readability of summaries by non-experts.

Concerning legal NLP, LEGAL-BERT demonstrated the efficiency of domain pre-training for legal texts understanding [10], LexGLUE created the standardized suite of benchmarks for legal NLP [11], and LexSumm/LexT5 proposed the legal summarization benchmark/model [12].Within Indian legal NLP, Bhattacharya et al. [13] ran the first systematic comparison of extractive and abstractive summarization algorithms on Indian Supreme Court judgments, a comparison later extended by Shukla et al. [14] across three legal summarization datasets; and Malik et al. [15] released ILDC, ≈35,000 Indian Supreme Court cases, for judgment prediction and explanation. These are valuable resources, but they focus on case law rather than the government policy and legislative documents addressed here.

On accessibility, Manor and Li [16] proposed summarizing legal contracts into plain English, showing that unsupervised extractive methods perform poorly given the stylistic gap between legal and plain language. This is the closest prior work to our accessibility layer, although it is limited to short contract clauses rather than full-length documents. Overall, existing work leaves gaps in summarizing lengthy Indian government and policy documents while preserving structure, and in optimizing summaries for non-expert readability. The proposed SET framework targets these gaps directly.

III. PROBLEM DEFINITION

Government and legal documents are central to governance and regulatory decision-making, but their length, technical terminology, and extensive cross-references make them difficult for non-expert readers to comprehend, so extracting key information remains challenging for automatic summarization systems.

Transformer models such as BART, PEGASUS, and T5 achieve strong abstractive summarization performance [3]–[5], but their fixed input length limits processing of long legal and government documents, and performance degrades as length increases. Most are also trained on general-domain news data, which differs from legal and policy text in structure, style, and terminology [1], [2], and even accurate summaries often retain complex legal terms and formal phrasing that are difficult for non-experts. This limitation is especially consequential for public policy documents intended for a broad audience. SET responds to these gaps by combining structured document processing with readability enhancement to produce concise, accessible summaries of long legal and government documents.

IV. METHODOLOGY

A. System Overview

SET is a hierarchical pipeline for long legal and government documents, dividing summarization into preprocessing, TF–IDF-based extraction, chunk-level abstractive summarization, post-processing, and accessibility enhancement (Fig. 1), so that fixed input-length limits of transformer models do not prevent processing of very long documents.

Preprocessing normalizes formatting inconsistencies, including broken paragraphs, headers, footnotes, page numbers, and encoding issues, that would otherwise degrade downstream processing. TF–IDF then scores sentences, combined with a position weight (since section-opening sentences often introduce key concepts) and a length weight (very short sentences tend to be labels; very long ones tend to hedge), producing a compact extract for abstractive summarization. Selected sentences are grouped into fixed-size chunks within BART's input limit and summarized independently; the chunk summaries are concatenated, deduplicated, and filtered for coherence. Finally, an accessibility layer simplifies legal terminology while preserving meaning. Each stage is modular and can be replaced independently.

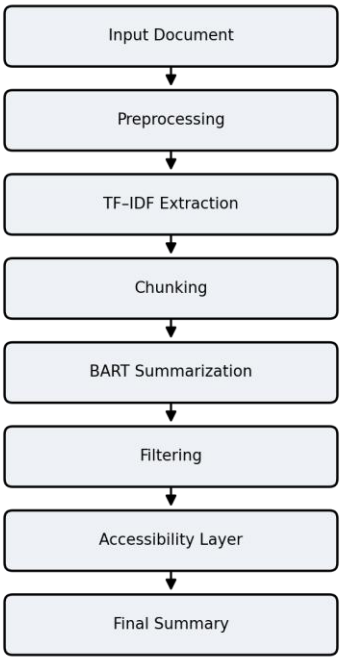


Fig. 1. SET hierarchical summarization pipeline.

B. Preprocessing Pipeline

Before extraction, documents are normalized and cleaned. Text originating from different government sources arrives in varied encodings (UTF-8, Latin-1, Windows-1252), so all input is first converted to UTF-8 and normalized using Unicode NFKC, which improves downstream tokenization. Repetitive elements such as headers, footers, footnotes, and table-of-contents entries are then removed through rule-based matching, and broken paragraphs or page-level artifacts are normalized by merging fragmented segments back into coherent blocks.

The cleaned text is split into sentences, which form the basic unit for TF–IDF scoring; sentences with a low proportion of alphabetic characters (for example, table fragments) are filtered out at this stage. Since BART-large-CNN accepts inputs of up to 1,024 tokens, the remaining sentences are grouped in order into fixed-size chunks that stay within this limit, and the

resulting sequence of document chunks forms the input to the hierarchical summarization pipeline described next.

C. TF–IDF Based Sentence Extraction

Pre-processed texts still have redundant preambles, cross-referencing, and definitions; salient sentence extraction before abstractive summarization saves costs and increases the importance of informative content by use of TF–IDF [9]:

$$TFIDF(s) = \Sigma t \in s\ tf(t,s) * \log(N/df(t)) \quad (8)$$

Position and length weight functions improve upon the above through a combined metric, $Score(s) = TFIDF(s) * Wpos * Wlen$ (9), where Wpos gives preference to opening sentences and Wlen discounts sentences based on length. The top-k sentences in each paragraph are selected, $Sselected = Top\text{-}k\{Score(si)\}$ (10), and provided to chunking.

D. Chunking Method

Selected sentences $Sselected = \{s1,\ldots, sn\}$ (11) are divided in order into chunks of size $Cj = \{sk,\ldots, sk+m\}$ of maximum size Lmax (12-13).Because tokens exceed words roughly as $Tokens \approx 1.3 \times Words$ (14), a chunk size of 700 words ($700 \times 1.3 \approx 910 < 1024$) (15) stays safely within BART's limit. Chunks that are predominantly non-alphabetic (tables, artifacts) are discarded.

This chunking strategy enables efficient summarization of long legal and government documents without exceeding the input-length limitations of transformer models. By preserving sentence order and coherent text segments, the generated summaries retain the overall semantic structure of the original document while remaining compatible with the BART-large-CNN architecture. The resulting chunks are passed to the abstractive summarization module, where each is summarized independently before being combined into the final document-level summary.

E. BART-Based Abstractive Summarization

Each chunk is summarized independently with BART-large-CNN [3], a denoising-autoencoder sequence-to-sequence transformer, $L = \log P(x|\tilde{x})$ (16), chosen for fluent, coherent generation. Chunk summaries $yj = BART(Cj)$ (17) are concatenated, $Y = \bigcup j\ yj$ (18) (Fig. 2), using identical decoding settings across chunks without additional fine-tuning.

Processing each document chunk independently keeps memory consumption low while preserving contextual information within each chunk, and enables the framework to summarize documents that far exceed the transformer's native input-length limit. Because BART is pretrained mainly on general-domain text rather than legal corpora, legal terminology and statutory references are not always represented with full accuracy. This limitation is addressed by the subsequent post-processing and accessibility stages.

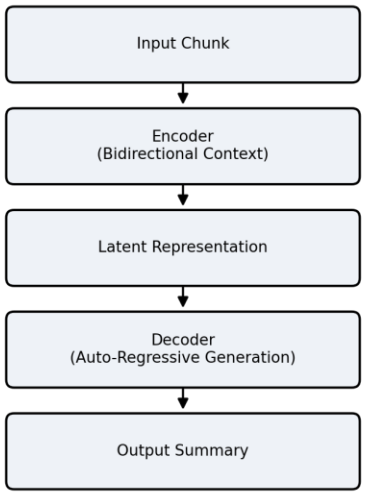


Fig. 2. Chunk-level abstractive summarization with BART.

F. Deduplication and Post-Processing

Because chunks are summarized independently, redundant content can appear across chunk summaries. Duplicate sentences are detected by comparing prefixes, $Duplicate(yi,yj) = 1$ if $prefixk(yi)=prefixk(yj)$, else 0 (19), keeping only the first occurrence. Rule-based filtering then removes incomplete fragments, redundant headers, repeated phrases, irrelevant content, and excessively short sentences before the remaining summaries are concatenated subject to a length cap, $Yfinal = \{y1,\ldots,ym\}$, $|Yfinal| \leq Lmax$ (20–21).

This rule-based design was chosen for its transparency, low computational overhead, and ease of integration with the hierarchical pipeline, removing redundant information while preserving semantic consistency across chunk-level summaries. The refined document summary is then passed to the accessibility enhancement module described next.

G. Accessibility Enhancement Layer

Even the output of a fluent transformer can contain formal phrasing with jargon that non-expert readers cannot understand; this in turn motivates this layer, which is similar in concept to the plain-language contract summarization task of Manor and Li [16]. Legal and government documents are meant for public distribution but the language used is generally in the legalese style , so even if summarized may still be hard for the general public to understand .
The sentences are re-ranked with the same composite score, $Score(s) = TFIDF(s) \times Wpos \times Wlen$ (22) Frequent legal expressions are replaced with a dictionary that has been manually curated where it does not make a difference in legal interpretation (e.g., shall→will; pursuant to→according to; notwithstanding→despite; hereinafter→from now on). Output is restructured into coherent paragraphs and bullet points where appropriate, improving presentation without changing semantic content, and readability is scored with the Flesch-Kincaid grade-level formula [17]:

$$FK = 206.835 - 1.015(W/S) - 84.6(Syl/W) \quad (23)$$

where W, S, and Syl are word, sentence, and syllable counts; higher FK indicates easier reading. Unlike systems that only shorten text, this layer targets readability and information preservation jointly. By combining legal-terminology simplification with structured formatting, it improves the usability of generated summaries without altering the rest of the pipeline, producing output that is concise, coherent, and easier for non-expert users to understand while preserving the essential information of the source document.

V. DATASET

There is no standard benchmark for long-document summarization of Indian legal and policy text. Thus, a custom corpus was created from official sources, India Code, National Portal of India, ministry websites, and departmental portals. All documents were retrieved from official government sources and were used solely for academic research purposes.

The dataset consists of 77 documents in 6 categories: Central Acts and Codes (28) e.g. CrPC, CGST Act, IT Act and related legislation, Constitutional Documents (8) e.g. Constitution of India and select amendments, State Bills and Acts (12), National Policy Reports (18) e.g. NEP 2020, National Water Policy, International Reports (6) e.g. WHO, UN, IPCC, and Annual Reports (5) e.g. RBI and other public institutions. Total: ≈2,785,594 words, average ≈36,176 words/document (range: ≈900 to >200,000 words), allows evaluation over a wide range of document length and structural complexity.

Source documents are linguistically complex with average Flesch−Kincaid scores of 15−25, reflecting formal legal language and domain-specific terminology. All 77 documents were manually inspected for integrity and compatibility with the preprocessing pipeline. 76 processed successfully without modification, one scanned, image-only document was excluded as reliable text extraction could not be performed. The data set is intentionally diverse in length, structure, and subject, but it is small compared to large-scale benchmarks such as ILDC [15]; we intend to extend it to other legal domains and Indian regional languages in future work.

VI. RESULTS AND ANALYSIS

A. ROUGE Evaluation

TABLE I. ROUGE VS. SOURCE DOCUMENTS

Method	R-1	R-2	R-L
Proposed SET	0.275	0.102	0.143
T5-small	0.210	0.071	0.120
Extractive Baseline	0.184	0.082	0.184

TABLE II. ROUGE VS. HUMAN REFERENCES

Document	R-1	R-2	R-L
NEP 2020	0.331	0.072	0.164
CrPC	0.237	0.040	0.130
CGST Act	0.283	0.074	0.155
NRHM Framework	0.294	0.061	0.132
J&K Reorg. Act	0.287	0.104	0.176
Average	0.286	0.070	0.151

SET (Table I) surpasses T5-small [5] and an extractive baseline on ROUGE-1/2 against source documents (the latter only wins ROUGE-L by directly copying source sentences). Compared with human references (Table II), SET performs best on NEP 2020 and worst on CrPC, showing the cross-referenced complexity of legal codes.

B. Readability and Human Evaluation

TABLE III. READABILITY (FLESCH–KINCAID) BY STAGE

Stage	FK Score	Level
Source Document	15–25	Very Difficult
BART Summary	34–40	Difficult
Accessible Output	50–60	Standard

TABLE IV. HUMAN EVALUATION (1–5 SCALE)

Document	Fluency	Coverage	Access.
NEP 2020	4.20	3.60	4.00
CrPC	3.20	2.60	2.60
CGST Act	2.40	2.00	2.60
NRHM Framework	4.00	3.60	4.00
J&K Reorg. Act	3.20	2.60	2.60
Average	3.40	2.88	3.24

The accessibility layer improves the overall readability by ≈30–40 FK points (Table III), with the greatest gain coming from that stage in particular. Human evaluation (Table IV) shows a consistent trend of fluency over content coverage; CGST Act scores the lowest, indicating that detailed statutory provisions are the most difficult to compress abstractively. The majority of summaries (54.5%) were 200-500 words in length, with 28.6% falling between 50-200 words and 16.9% under 50 words; no summaries exceeded 500 words.

C. Key Observations

- SET consistently outperforms T5-small on ROUGE-1/2.
- Policy documents are more effective in summarizing than legal codes because there are fewer cross-references.
- The accessibility layer is more readable, but doesn't add much to the summaries.
- Fluency is rated higher than content coverage in human ratings – future work should target information preservation

VII. DISCUSSION AND ERROR ANALYSIS

The FK improvement from 15-25 to 50-60 confirms the efficacy of the accessibility layer but further exploration is needed regarding how much simplified summaries preserve legal meaning. The CGST Act is the most challenging case, where BART's fluent output sometimes overgeneralizes some conditional statutory provisions, indicating a limited exposure to the legal domain during its pre-training.

Chunking is effective for handling input-length constraints, but it can lose context across chunk boundaries, especially for documents with long-range cross-references. This shows up in several recurring failure modes. Domain-specific language suffers most visibly: conditional provisions, such as those in the CGST Act, are sometimes oversimplified in ways that reduce legal precision. Global context can also be lost, since definitions introduced early in a document may fall outside the window of a later chunk, leading to inconsistent terminology across the summary. Prefix-based deduplication misses redundancy that is semantically equivalent but expressed in different words, and explanatory sentences with low TF-IDF scores are sometimes omitted altogether, leaving some provisions without their surrounding context. Finally, expanding the terminology dictionary used by the accessibility layer requires legal domain expertise, since substitutions must be made safely without altering legal meaning. These trade-offs between the efficiency gained from extractive filtering and chunking, and the contextual continuity lost in the process, explain why policy documents such as NEP 2020 summarize better than tightly cross-referenced legal codes such as the CGST Act, and motivate document-specific strategies as future work.

VIII. CONCLUSION

In this paper, we proposed SET, a hierarchical framework for long Indian legal and government documents that consists of TF-IDF extraction, BART summarization at the chunk level, post-processing and readability enhancement. In a curated 77-document corpus, SET achieved competitive ROUGE performance while increasing Flesch–Kincaid readability from 15–25 to 50–60. Human evaluation confirmed fluent, coherent, accessible output. Documents with dense cross-referencing (e.g., the CGST Act) remain more challenging to distill accurately than policy-oriented documents. Nonetheless, the framework offers an effective, computationally efficient approach to improving public access to legal and government information on CPU-only systems.

IX. FUTURE WORK

The priorities include fine-tuning on domain-specific legal corpora, expansion of the terminology dictionary with expert validation, adoption of long-context architectures such as Longformer [6] and LongT5 [8] to minimize cross-chunk context loss, support for Indian regional languages, incorporation of retrieval-augmented generation and legal LLMs to improve factual consistency, extension of resources such as ILDC [15] with gold-standard summaries for stronger evaluation, and deployment of the framework as a web application for real-time summarization of newly published government notifications.

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